**Advanced Data Analysis, SQL Joins & Business KPI Dashboard**

**1. Introduction**

This project focuses on using SQL, Microsoft Excel, and Power BI to analyze Superstore sales data and develop meaningful business insights. The objective is to strengthen data analysis skills, calculate business KPIs, and present findings through clear visual dashboards.

**2. Objectives**

- Perform sales analysis using SQL queries.
- Apply aggregation, grouping, filtering, and sorting techniques.
- Analyze sales by category, sub-category, and region.
- Calculate relevant business KPIs where the required data is available.
- Create an interactive Power BI dashboard.
- Present business findings in a clear and professional format.

**3. Tools and Technologies**

- PostgreSQL — SQL querying and data analysis
- Microsoft Excel — data organization, PivotTables, and KPI calculations
- Microsoft Power BI — dashboard creation and visualization

**4. SQL Analysis**

SQL queries were prepared to analyze sales by category, region, and sub-category. Aggregate functions such as SUM(), along with GROUP BY and ORDER BY, were used to summarize and rank sales performance.

Profit and profit-margin analysis can be included after verifying that the relevant columns contain valid numeric values. JOIN analysis requires suitable related tables or matching key columns.

**5. Excel Analysis**

Excel was used to summarize sales data, calculate available KPIs, and prepare PivotTables for category, region, segment, and customer analysis. Monthly sales trends can also be analyzed when a valid order-date field is available.

**6. Power BI Dashboard**

The planned dashboard includes:

- Total Sales and other verified KPIs
- Category-wise Sales
- Region-wise Sales
- Sub-category-wise Sales
- Monthly Sales Trend, if order dates are available

- Interactive Region and Category slicers

**7. Business Insights**

The analysis can help identify high-performing product categories, regions with strong sales, and sub-categories that contribute significantly to revenue. These findings can support inventory planning, sales strategy, and business decision-making.

Final insights should be added after checking the actual query results and dashboard values.

**8. Testing and Validation**

- Verify the dataset's row count and column names.
- Check for missing, duplicate, or incorrectly formatted values.
- Validate SQL query results against Excel summaries.
- Confirm that Power BI visuals display the correct totals.
- Test dashboard slicers and filters.

**9. Conclusion**

This task provides practical experience in SQL-based business analysis, Excel reporting, KPI development, and Power BI dashboard design. It demonstrates how structured data analysis can transform sales records into useful business information.